Gaussian Model

This section applies to the cells (i, ℓ) with `model` = `gaussian`. Inside it, the subscript $i\ell$ is dropped from the per-cell parameters and from the states; k is the day.

Model. A mission increment is Gaussian, $\mathcal{N}(\mu_{jk}^{\text{inc}}, v_{jk}^{\text{inc}})$ (inputs `mu` and `v`); the initial damage is $\mathcal{N}(\mu_0, v_0)$ and each replacement draw is $\mathcal{N}(\mu^{\text{new}}, v^{\text{new}})$. A known damage has variance 0.
Assumption N1 (independence). The increments, the initial damage and the replacement draws are mutually independent. A repair acts on the damage path by the law of the cell (Section **??**). The Gaussian law gives negative damage a small probability; the input checks of Section **??** ($\mu \geq 3\sqrt{v}$ for the initial damage and for each increment) keep it at most $\Phi(-3) = 0.135\,\%$ for each of these terms.

Tracked states and repair laws. The cell tracks the mean μ_k and the variance v_k, and under ARD_1 their anchors g_k^{μ} and g_k^{v}. Both laws are accepted. The exact day maps are

$$\mu_k = \begin{cases} \mu_{k-1} + \delta\mu_k & \text{no intervention on day } k, \\ (1-\rho)\,\mu_{k-1} & \text{repair, } \text{ARD}_\infty, \\ (1-\rho)\,\mu_{k-1} + \rho\, g_{k-1}^{\mu} & \text{repair, } \text{ARD}_1, \\ \mu^{\text{new}} & \text{replacement}, \end{cases} \tag{1}$$

$$v_k = \begin{cases} v_{k-1} + \delta v_k & \text{no intervention on day } k, \\ (1-\rho)^2\, v_{k-1} & \text{repair, } \text{ARD}_\infty, \\ (1-\rho)^2\, v_{k-1} + \big(1-(1-\rho)^2\big)\, g_{k-1}^{v} & \text{repair, } \text{ARD}_1, \\ v^{\text{new}} & \text{replacement}, \end{cases} \tag{2}$$

with $\delta\mu_k = \sum_{j=1}^{M} x_{ijk}\, \mu_{jk}^{\text{inc}}$ and $\delta v_k = \sum_{j=1}^{M} x_{ijk}\, v_{jk}^{\text{inc}}$.
Proposition 0.1 (The state stays Gaussian)**.** *Under N1, on every day of a schedule $D_k \sim \mathcal{N}(m_k, V_k)$, where (m_k, V_k) follow the maps (1) and (2), with anchors equal to the mean and the variance just after the last intervention.*

Proof. A mission adds an independent Gaussian term, and ARD_∞ scales the damage by $1-\rho$; both keep the law Gaussian and give the first two cases. Under ARD_1, $D^- = G + \Delta$, with G the damage just after the last intervention and Δ the increments added since; G and Δ are independent Gaussian variables by N1. Then $D^+ = G + (1-\rho)\Delta$ is Gaussian, with mean $m_G + (1-\rho)(m^- - m_G)$ and variance

$$V_G + (1-\rho)^2\, V_\Delta = V_G + (1-\rho)^2\,(V^- - V_G) = (1-\rho)^2\, V^- + \big(1-(1-\rho)^2\big)\, V_G.$$

A replacement starts from the replacement draw. Therefore the law is Gaussian on every day, with the moments of the maps. □

So ARD_1 is exact for this model. Its variance map uses the factor $(1-\rho)^2$ and the variance anchor g^v. The rows are the generic rows of Section **??**, with the data below; $V_{\max} = \tau^2/\kappa_G^2$, with $\kappa_G = \Phi^{-1}(1-\varepsilon)$.

State	Day increment	Factor a	Natural range	Initial value	Replacement value
μ; anchor g^{μ}	$\delta\mu_k$	$1-\rho$	τ	μ_0	μ^{new}
v; anchor g^{v}	δv_k	$(1-\rho)^2$	$V_{\max}$	v_0	v^{new}

Reliability row

Gaussian cells need $\varepsilon < 0.5$, so that $\kappa_G = \Phi^{-1}(1-\varepsilon) > 0$. For $D \sim \mathcal{N}(m, V)$, $\Pr(D > \tau) = 1 - \Phi\big((\tau - m)/\sqrt{V}\big) \le \varepsilon$ exactly when $m + \kappa_G\sqrt{V} \le \tau$. With $\kappa_G > 0$ and $\tau - \mu \ge 0$ this is equivalent to the squared row, imposed on every day $k = 1, \dots, T$:

$$\kappa_G^2\, v_k \le (\tau - \mu_k)^2, \qquad \mu_k \le \tau. \tag{3}$$

Validity. Let $(m_k, V_k) \le (\mu_k, v_k)$ be the true moments (Proposition 0.1 and Section **??**). If $V_k = 0$, then $D_k = m_k \le \tau$. Otherwise $\mu_k < \tau$, because $\mu_k = \tau$ forces $v_k = 0$, and $\Pr(D_k > \tau) = 1 - \Phi\big((\tau - m_k)/\sqrt{V_k}\big) \le 1 - \Phi\big((\tau - \mu_k)/\sqrt{v_k}\big) \le \varepsilon$, because $(\tau - \mu)/\sqrt{v}$ decreases in μ and in v.

Forms. In the form `exact`, Eq. (3) is a *nonconvex* quadratic constraint: $(\mu, v) = (\tau, 0)$ and $(0, V_{\max})$ satisfy it, and their midpoint $(\tau/2, V_{\max}/2)$ does not, since $\tau^2/2 > \tau^2/4$. The solver handles it through Gurobi's nonconvex quadratic support (parameter `NonConvex` = 2). The row has the form $v_k \le f(\mu_k)$ with the convex function

$$f(\mu) = \frac{(\tau - \mu)^2}{\kappa_G^2}, \qquad \text{side row } \mu \le \tau,$$

so the forms `tangent` and `pwl` of Section **??** apply, and both keep the side row.

Remark (the level ε and the range $V_{\max}$). $V_{\max} = f(0) = \tau^2/\kappa_G^2$ is the largest variance that the row admits, at $\mu = 0$, so it is the natural range of v and the big-M constant of its rows; the input checks of Section **??** keep v_0, its starting anchor and v^{new} inside it. $V_{\max}/\tau^2$ is 0.18 at $\varepsilon = 0.01$, 0.61 at 0.1, 3.6 at 0.3 and about 1600 at 0.49. It grows without bound as $\varepsilon \to 0.5$, and at $\varepsilon \ge 0.5$ the squared row does not describe the reliability requirement, because $\kappa_G \le 0$. Hence the condition $\varepsilon < 0.5$ (Section **??**); the range is large only close to 0.5.

Optional row. If `v_max_user` is given for the cell, the row $v_k \le V^{\text{user}}$ is added on every day.

Facts used by the encoding and loop arguments. (M1) The day maps are nondecreasing in (μ, v, g^μ, g^v): their coefficients lie in $[0, 1]$. (M2) The true moments follow the maps exactly (Proposition 0.1). (M3) If the tracked states satisfy Eq. (3) and the true moments are not larger, then $\Pr(D_k > \tau) \le \varepsilon$ (the validity paragraph). The loop row of Section **??** covers μ, v and, under ARD_1, g^μ and g^v.

Inverse Gaussian Model

This section applies to the cells (i, ℓ) with `model` = `inverse_gaussian` (IG). Inside it, the subscript $i\ell$ is dropped from the per-cell parameters and from the states; k is the day.

Model. $\text{IG}(\mu, \lambda)$ is the inverse Gaussian law with mean $\mu > 0$ and shape $\lambda > 0$. Its density is

$$f(x; \mu, \lambda) = \sqrt{\frac{\lambda}{2\pi x^3}}\, \exp\Big(-\frac{\lambda\,(x-\mu)^2}{2\mu^2 x}\Big), \qquad x > 0,$$

and its variance is μ^3/λ. For $c > 0$, $c\,\text{IG}(\mu, \lambda) = \text{IG}(c\mu, c\lambda)$. The cell has one dispersion $\eta > 0$ (input `eta`), and a term of mean a has the law $\text{IG}(a, a/\eta) = a\,Z$ with $Z \sim \text{IG}(1, 1/\eta)$. Its density is $\sqrt{a/(2\pi\eta x^3)}\, \exp\big(-(x-a)^2/(2\eta a x)\big)$, and η is its squared coefficient of variation. Every term has this law with its own mean: a mission increment (mean μ_{jk}^{inc}), the initial damage (μ_0) and each replacement draw (μ^{new}). A known damage a is the term $a \cdot 1$. The terms may depend on each other in any way: this model needs no independence assumption.

Tracked state and repair laws. The cell tracks the mean μ_k, and under ARD_1 its anchor g^{μ}_k. Both laws are accepted. The exact day map of the mean is

$$\mu_k = \begin{cases} \mu_{k-1} + \delta\mu_k & \text{no intervention on day } k, \\ (1-\rho)\,\mu_{k-1} & \text{repair, } \mathrm{ARD}_\infty, \\ (1-\rho)\,\mu_{k-1} + \rho\, g^{\mu}_{k-1} & \text{repair, } \mathrm{ARD}_1, \\ \mu^{\mathrm{new}} & \text{replacement}, \end{cases} \tag{4}$$

with $\delta\mu_k = \sum_{j=1}^{M} x_{ijk}\,\mu^{\mathrm{inc}}_{jk}$. The rows are the generic rows of Section **??**, with the data below.

State	Day increment	Factor a	Natural range	Initial value	Replacement value
μ; anchor g^{μ}	$\delta\mu_k$	$1-\rho$	τ	μ_0	μ^{new}

Reliability row

Between replacements, the damage on day k is $D_k = \sum_t w_t X_t = \sum_t (w_t a_t)\, Z_t$, where $X_t = a_t Z_t$ are the terms added since the last replacement (or day 0), and each Z_t has the law of Z or equals 1. Each weight $w_t \in (0,1]$ is a product of factors $1-\rho$: ARD_∞ multiplies every weight by $1-\rho$, and ARD_1, with $D^{+} = G + (1-\rho)\,\Delta$, multiplies only the weights of the terms added since the last intervention. When the starting anchor g^{μ}_0 is below μ_0, the initial damage $\mu_0 Z_0$ is read as the two terms $g^{\mu}_0 Z_0$ and $(\mu_0 - g^{\mu}_0) Z_0$. The mean $\sum_t w_t a_t$ is at most the tracked mean μ_k (Section **??**).

Proposition 0.2 (Stop-loss order)**.** *Let $D = \sum_t w_t a_t Z_t$ with repair weights $w_t \in (0,1]$, means $a_t \geq 0$, $\sum_t w_t a_t \leq \mu$, and each Z_t either with the law of Z or equal to 1, under any joint law. Then $\mathbb{E}(D-d)_+ \leq \mathbb{E}(\mu Z - d)_+$ for every $d \geq 0$.*

Proof. Let $c_t = w_t a_t$ and $A = \sum_t c_t$. If $A = 0$, then $D = 0$. Otherwise, $x \mapsto (x-d)_+$ is convex and nondecreasing, $Z \geq 0$, and $(A-d)_+ \leq \mathbb{E}(AZ-d)_+$ by Jensen's inequality ($\mathbb{E}Z = 1$), so

$$\mathbb{E}(D-d)_+ = \mathbb{E}\Big(\sum_t \frac{c_t}{A}\,(AZ_t) - d\Big)_+ \leq \sum_t \frac{c_t}{A}\,\mathbb{E}(AZ_t - d)_+ \leq \mathbb{E}(AZ-d)_+ \leq \mathbb{E}(\mu Z - d)_+.$$

□

Let q be the $(1-\varepsilon)$-quantile of Z, $\Pr(Z > q) = \varepsilon$, and let $C = \mathbb{E}[Z \mid Z > q]$, the conditional value at risk (CVaR) of Z at the level $1-\varepsilon$. The cap and the row, on every day $k = 1, \ldots, T$, are

$$\bar{\mu} = \frac{\tau}{C}, \qquad \mu_k \leq \bar{\mu}. \tag{5}$$

Theorem 0.3 (CVaR cap)**.** *Let $D \geq 0$ satisfy $\mathbb{E}(D-d)_+ \leq \mathbb{E}(\mu Z - d)_+$ for every $d \geq 0$, with $0 \leq \mu \leq \bar{\mu}$. Then $\Pr(D > \tau) \leq \varepsilon$.*

Proof. If $\mu = 0$, then $\mathbb{E}D \leq 0$ and $D = 0$. Let $\mu > 0$ and $d = \mu q$. Since $C > q$, $d < \mu C \leq \tau$. For every x, $\mathbf{1}\{x > \tau\} \leq (x-d)_+/(\tau - d)$, and $\mathbb{E}(Z-q)_+ = \varepsilon\,(C-q)$. Then

$$\Pr(D > \tau) \leq \frac{\mathbb{E}(D-d)_+}{\tau - d} \leq \frac{\mu\,\mathbb{E}(Z-q)_+}{\tau - \mu q} = \frac{\varepsilon\,\mu\,(C-q)}{\tau - \mu q} \leq \varepsilon,$$

where the last step is $\mu C \leq \tau$. □

By Proposition 0.2 with $\mu = \mu_k$ and Theorem 0.3, the row (5) gives $\Pr(D_k > \tau) \le \varepsilon$ on every day, under both laws. The row is linear: it is built in its exact form, and a value of `reliability_impl` at the cell is ignored (Section **??**).

Computation of $\bar{\mu}$ (offline, once per cell, before the model is built). With $r(x) = 1/\sqrt{\eta x}$, the distribution function of Z is

$$F(x) = \Phi\big(r(x)\,(x-1)\big) + e^{2/\eta}\,\Phi\big(-r(x)\,(x+1)\big).$$

The second term overflows for small η; it is computed as $\exp\big(2/\eta + \ln\Phi(-r(x)(x+1))\big)$, with $\ln\Phi$ from a log-space routine (`scipy.special.log_ndtr`).

1. q is the root of $1 - F(q) = \varepsilon$, found by bisection or by `brentq` on $\ln q$. The method `isf` of `scipy.stats.invgauss`, with `mu` $= \eta$ and `scale` $= 1/\eta$, gives the same value.
2. The tail mean has the closed form $\mathbb{E}[Z;\, Z > q] = F(1/q)$, so $\bar{\mu} = \tau\,\varepsilon/F(1/q)$. Derivation: substituting into the density gives $z\,f_Z(z) = z^{-2} f_Z(1/z)$, and the change of variable $u = 1/z$ gives $\int_q^\infty z\,f_Z(z)\,dz = \int_0^{1/q} f_Z(u)\,du = F(1/q)$.

Values of $\bar{\mu}/\tau$:

	$\eta = 0.05$	0.25	1	4	10
$\varepsilon = 0.001$	0.491	0.242	0.101	0.0370	0.0190
$\varepsilon = 0.01$	0.570	0.322	0.156	0.0679	0.0401
$\varepsilon = 0.1$	0.693	0.477	0.299	0.183	0.141

Remark (the largest cap for this order). Let $\mu > \bar{\mu}$. For a tail probability $\varepsilon' > \varepsilon$ close to ε, with $(1-\varepsilon')$-quantile q', still $\mu\,\mathbb{E}[Z \mid Z > q'] > \tau$. The two-point law $D = \mu\,\mathbb{E}\big[Z \mid \mathbf{1}\{Z > q'\}\big]$ is below μZ in the stop-loss order, by Jensen's inequality, and $\Pr(D > \tau) \ge \varepsilon'$. So no larger cap is safe for every law that Proposition 0.2 allows. That class also contains laws that no schedule produces, so the row can reject some safe schedules.

Remark (the quantile cap is not the row). The cap τ/q makes the model law μZ meet the target exactly: $\Pr(\mu Z > \tau) = \varepsilon$ at $\mu = \tau/q$. A sum of terms below μZ in the stop-loss order can have a larger tail. Example: $\eta = 10$, $\varepsilon = 0.1$, so $\tau/q = 0.473\,\tau$; three independent missions of equal mean, with total mean τ/q, give $\Pr(D > \tau) = 0.118$. The CVaR cap is $0.141\,\tau$ there. At $\varepsilon = 0.01$ and η from 0.05 to 10, $\bar{\mu}$ is 0.93 to 0.59 times τ/q.

Facts used by the encoding and loop arguments. (M1) The day map (4) is nondecreasing in (μ, g^μ). (M2) The true mean follows it exactly. (M3) If $\mu_k \le \bar{\mu}$ and the true mean is not larger, then $\Pr(D_k > \tau) \le \varepsilon$ (Proposition 0.2, Theorem 0.3). The loop row of Section **??** covers μ and, under ARD_1, g^μ. Each day of each repetition has its own set of terms; the argument above holds for each of them, because it needs neither independence nor a fixed set of terms.

Wiener Model

This section applies to the cells (i, ℓ) with `model` $=$ `wiener`. Inside it, the subscript $i\ell$ is dropped from the per-cell parameters and from the states; k is the day.

Model. The damage is a Wiener process with drift, observed once a day. A mission day adds a Gaussian increment $\mathcal{N}(\mu_{jk}^{\mathrm{inc}}, \sigma^2)$: the mean comes from the input `mu`, and the diffusion constant $\sigma > 0$ (input `sigma`) is the same for every mission, because it describes the material of the component, not the route. The initial damage is $\mathcal{N}(\mu_0, v_0)$ with $v_0 \ge 0$, and each replacement draw is $\mathcal{N}(\mu^{\mathrm{new}}, v^{\mathrm{new}})$. **Assumption W1 (independence).** These terms are mutually independent.

Repairs act on the mean only (a modelling choice). A repair shifts the damage by a known amount: $D^+ = D^- - \rho\, m^-$ under ARD_∞ and $D^+ = D^- - \rho\,(m^- - g^\mu)$ under ARD_1, with $m^- = \mathbb{E}D^-$. The law stays Gaussian, the variance is unchanged, and only a replacement lowers it.

Tracked states and repair laws. The cell tracks μ_k and v_k, and under ARD_1 the mean anchor g_k^μ; the variance has no anchor. Both laws are accepted. The mean follows the map (1) of the Gaussian model. The variance map is

$$v_k = \begin{cases} v_{k-1} + \delta v_k & \text{no intervention on day } k, \\ v_{k-1} & \text{repair, both laws,} \\ v^{\mathrm{new}} & \text{replacement,} \end{cases} \qquad \delta v_k = \sigma^2 \sum_{j=1}^{M} x_{ijk}. \tag{6}$$

Under W1 the damage is $\mathcal{N}(m_k, V_k)$ on every day, with (m_k, V_k) given by these maps: a mission adds an independent Gaussian term, and a repair or a replacement acts as stated above.

The rows are the generic rows of Section **??**, with the data below. The variance takes the repair map $\psi = v_{k-1}$, that is the factor $a = 1$ under both laws, and no anchor. A depot day adds no increment, so for v the generic no-intervention and repair rows together equal the single row

$$v_k \geq v_{k-1} + \delta v_k - V_{\max}\, x_k^r, \tag{7}$$

which may replace them in the big-M form.

State	Day increment	Factor a	Natural range	Initial value	Replacement value
μ; anchor g^μ	$\delta\mu_k$	$1-\rho$	τ	μ_0	μ^{new}
v; no anchor	δv_k	1	$V_{\max}$	v_0	v^{new}

Remark (a repair of the path). A repair of the damage path, as in the Gaussian model, gives the same mean and the variance $(1-\rho)^2 V$ under ARD_∞, or $(1-\rho)^2 V + (1-(1-\rho)^2)\, V_G$ under ARD_1, with $V_G \leq V$ the variance just after the last intervention. Both are at most V. The reliability row increases in v, so the rows of this section are safe for that reading as well.

Reliability row

The row is the Gaussian row (3), with $\kappa_G = \Phi^{-1}(1-\varepsilon)$ and $\varepsilon < 0.5$, on every day $k = 1, \ldots, T$; its validity argument, its forms (`exact`, `tangent`, `pwl`, with the same function f and side row) and the remark on $V_{\max} = \tau^2/\kappa_G^2$ are those of Section 0.1. If `v_max_user` is given for the cell, the row $v_k \leq V^{\mathrm{user}}$ is added on every day.

Facts used by the encoding and loop arguments. (M1) The day maps are nondecreasing in (μ, v, g^μ). (M2) The true moments follow them exactly. (M3) As for the Gaussian model. The loop row of Section **??** covers μ, v and, under ARD_1, g^μ.

Remark (the loop forces a replacement). The variance grows by σ^2 on every mission day and falls only at a replacement. If the train flies n missions in the operating phase and the component is not replaced there, then $v_T \geq v_{H_1} + n\,\sigma^2 > v_{H_1}$, and the loop row $v_T \leq v_{H_1}$ fails. So every feasible schedule includes a replacement of the component in the operating phase whenever its train flies a mission there. No input check enforces this. With `allow_replacement` false, the train of a Wiener cell cannot fly in the operating phase.

Gamma Model

This section applies to the cells (i, ℓ) with `model` = `gamma`. Inside it, the subscript $i\ell$ is dropped from the per-cell parameters and from the states; k is the day.

Model. Gamma(a, β) is the Gamma law with shape $a > 0$ and rate $\beta > 0$; its mean is a/β. Every term that adds damage is a Gamma variable with its own rate:

- a mission increment with mean $\mu_{jk}^{\text{inc}} > 0$ (input `mu`) and rate β_{jk} (inputs `gamma_beta` and `gamma_beta_trans`) is Gamma$(\beta_{jk}\,\mu_{jk}^{\text{inc}},\ \beta_{jk})$;
- the initial damage with mean $\mu_0 > 0$ is Gamma$(\beta_0\,\mu_0,\ \beta_0)$ when its rate β_0 is given (input `gamma_beta_0`); when `gamma_beta_0` is omitted, the initial damage is known: the constant μ_0;
- each replacement draw with mean $\mu^{\text{new}} > 0$ is Gamma$(\beta^{\text{new}}\mu^{\text{new}},\ \beta^{\text{new}})$ when its rate β^{new} is given (input `gamma_beta_new`); when `gamma_beta_new` is omitted, it is the known constant μ^{new}.

Assumption G1 (independence). All these terms are mutually independent.

A repair acts on the damage path by the law of the cell (Section **??**): ARD_∞ gives $D^+ = (1-\rho)\,D^-$, and ARD_1 gives $D^+ = D^{\text{last}} + (1-\rho)\,(D^- - D^{\text{last}})$. After a repair the damage is, in general, not a Gamma variable: it is a weighted sum of independent Gamma terms and known constants. The reliability row below holds for every such sum.

Tracked state and repair laws. The cell tracks the mean μ_k only, and under ARD_1 its anchor g_k^μ. Both laws are accepted. The shape of the damage is not tracked. The exact day map of the mean is

$$\mu_k = \begin{cases} \mu_{k-1} + \delta\mu_k & \text{no intervention on day } k, \\ (1-\rho)\,\mu_{k-1} & \text{repair, } \text{ARD}_\infty, \\ (1-\rho)\,\mu_{k-1} + \rho\, g_{k-1}^\mu & \text{repair, } \text{ARD}_1, \\ \mu^{\text{new}} & \text{replacement}, \end{cases} \tag{8}$$

with $\delta\mu_k = \sum_{j=1}^{M} x_{ijk}\,\mu_{jk}^{\text{inc}}$. The map is exact for the true mean $\mathbb{E}D_k$, because expectation is linear. The rows are the generic rows of Section **??**, with the data below.

State	Day increment	Factor a	Natural range	Initial value	Replacement value
μ; anchor g^μ	$\delta\mu_k$	$1-\rho$	τ	μ_0	μ^{new}

Smallest rate. The cell's smallest rate is

$$\beta^* = \min\Big(\{\beta_{jk} : \mu_{jk}^{\text{inc}} > 0\} \cup \{\beta_0 : \mu_0 > 0\} \cup \{\beta^{\text{new}} : \mu^{\text{new}} > 0\}\Big), \tag{9}$$

where j runs over the missions and k over the days of both phases, and β_0, β^{new} enter only when they are given. So β^* is the smallest rate among the terms with a positive mean: a mission with $\mu_{jk}^{\text{inc}} = 0$ adds no term, and an omitted rate declares a known damage, which is a constant and does not enter β^*. If no term has a rate, the damage is a known constant, and the row below is $\mu_k \le \tau$. Write $t = \beta^*\tau$. Units of $1/\beta^*$ turn every law below into a law of rate at least 1.

Class of true states

Definition 0.4 (Class of weighted Gamma sums)**.** Let $m \ge 0$. The class $\mathcal{C}(m)$ holds the laws of the random variables

$$S = \delta + \sum_{i=1}^{n} c_i\, Y_i, \qquad Y_i \sim \text{Gamma}(a_i, 1) \text{ independent}, \quad c_i \in (0, 1], \quad a_i > 0, \quad \delta \ge 0,$$

with $n \geq 0$ and mean $\delta + \sum_i c_i a_i = m$. The term $c\,Y$ with $Y \sim \text{Gamma}(m'/c, 1)$ has mean m'; as $c \to 0$ it tends to the constant m'.

Lemma 0.5 (True states belong to the class)**.** *Let D_k be the damage of a Gamma cell on day k of a schedule, under either law. Then the law of $\beta^* D_k$ belongs to $\mathcal{C}(\beta^* \mathbb{E}D_k)$.*

Proof. Since the last replacement (or day 0), the damage is $D_k = \delta + \sum_t w_t X_t$: the X_t are the independent Gamma terms added since then, $\delta \geq 0$ is the weighted sum of the known constants, and each weight $w_t \in (0, 1]$ is a product of factors $1 - \rho$. A mission adds a term with weight 1. ARD_∞ multiplies every weight, constants included, by $1 - \rho$. ARD_1 multiplies by $1 - \rho$ the weights of the terms and constants added since the last intervention, and keeps the others. A replacement starts a new sum with the replacement draw. When the starting anchor g_0^μ is below μ_0, the initial damage is read as two independent terms of rate β_0, with means g_0^μ and $\mu_0 - g_0^\mu$, whose sum has the law $\text{Gamma}(\beta_0\mu_0, \beta_0)$; a known initial damage is read as the two constants g_0^μ and $\mu_0 - g_0^\mu$. Write $X_t = Y_t/\beta_t$ with $Y_t \sim \text{Gamma}(a_t, 1)$ and $\beta_t \geq \beta^*$ by Eq. (9). Then

$$\beta^* D_k = \beta^* \delta + \sum_t \frac{w_t\,\beta^*}{\beta_t}\, Y_t, \qquad \frac{w_t\,\beta^*}{\beta_t} \in (0, 1].$$

The Y_t are independent by Assumption G1. Therefore the law of $\beta^* D_k$ is in $\mathcal{C}(\beta^* \mathbb{E}D_k)$. □

The largest tail at a given mean

For $0 < m < t$, let $W(m, t)$ be the supremum of $\Pr(S > t)$ over the random variables S whose law is in $\mathcal{C}(m)$. $Q(a, x) = \Pr(\text{Gamma}(a, 1) > x)$ is the regularised upper incomplete gamma function (`scipy.special.gammaincc(a, x)`). The one-term member $c\,\text{Gamma}(m/c, 1)$ has the tail $Q(m/c,\, t/c)$ at t.

A member of $\mathcal{C}(m)$ is described by its atoms: the pairs (m_i, c_i) of a term mean $m_i = c_i a_i$ and a scale c_i, and the constant δ, read as an atom at scale $c = 0$. Terms with equal scales merge into one term, because a sum of independent Gamma variables of one rate is a Gamma variable.

Lemma 0.6 (First-order rates)**.** *Let S be a member with at least one term, with density f_S. Since $\delta \leq m < t$, f_S is continuous at t. For $c \in (0, 1]$ let*

$$\varphi_S(c) = \frac{1}{c} \int_0^\infty \Pr(t - y < S \leq t)\, \frac{e^{-y/c}}{y}\, dy, \qquad \varphi_S(0) = f_S(t).$$

(a) Adding mean η at scale c (a term $c\,\text{Gamma}(\eta/c, 1)$, or the constant η when $c = 0$) raises $\Pr(S > t)$ at the rate $\varphi_S(c)$ as $\eta \to 0$; removing mean from an atom at scale c lowers it at the same rate. (b) Moving the scale of an atom (m_i, c_i), $c_i \in (0, 1)$, at fixed mean m_i changes $\Pr(S > t)$ at the rate $m_i\, \partial\varphi_S/\partial c\,(c_i)$.

Proof. S is infinitely divisible, with drift δ and Lévy density $\ell(y) = \sum_i (m_i/c_i)\, e^{-y/c_i}/y$ on $y > 0$. A change of the parameters at the rates $(\dot\delta, \dot\ell)$ changes the Laplace transform $\mathbb{E}e^{-\sigma S} = \exp\bigl(-\sigma\delta - \int_0^\infty (1 - e^{-\sigma y})\, \ell(y)\, dy\bigr)$ at the rate $\int_0^\infty \mathbb{E}\bigl[e^{-\sigma(S+y)} - e^{-\sigma S}\bigr]\, \dot\ell(y)\, dy - \sigma\dot\delta\, \mathbb{E}e^{-\sigma S}$. This is the transform of the rate $\int_0^\infty \Pr(t - y < S \leq t)\, \dot\ell(y)\, dy + \dot\delta\, f_S(t)$ of $\Pr(S > t)$. Case (a) is $\dot\ell(y) = c^{-1} e^{-y/c}/y$, or $\dot\delta = 1$ when $c = 0$. Case (b) is $\dot\ell(y) = m_i\, \partial_c\bigl(c^{-1} e^{-y/c}/y\bigr)$ at $c = c_i$. □

Theorem 0.7 (One term is the worst case)**.** *Let $0 < m < t$. Then*

$$W(m, t) = \max_{c \in (0,1]} Q\left(\frac{m}{c},\, \frac{t}{c}\right). \tag{10}$$

Proof. Let $\mathcal{C}_K(m)$ be the members with at most K atoms, the constant included, and let $K \geq 3$.

1. *A maximiser exists.* $\mathcal{C}_K(m)$ is the image of the compact set of atoms (m_i, c_i) with $m_i \geq 0$, $\sum_i m_i = m$, $c_i \in [0,1]$. The law of S depends continuously on the atoms, and only the constant member $S = m$ has an atom, at $m \neq t$. So $\Pr(S > t)$ is continuous and attains its maximum at some S^*. Since $\Pr(m > t) = 0 < Q(m,t)$, S^* has at least one term. Every member has finitely many atoms, so $W(m,t) = \sup_K \max_{\mathcal{C}_K(m)} \Pr(S > t)$.

2. *First-order conditions.* Write $\varphi^* = \varphi_{S^*}$. By Lemma 0.6: (i) moving mean between two atoms of positive mean must not raise the tail, so φ^* takes one value κ^* at every atom of positive mean; (ii) at an atom with $c \in (0,1)$, $\partial\varphi^*/\partial c = 0$; (iii) if S^* has fewer than K atoms of positive mean, a new atom can be opened at any $c \in [0,1]$, so $\varphi^*(c) \leq \kappa^*$ on $[0,1]$.

3. *A single-peak function.* Put $\lambda = 1/c$ and $h(u) = \Pr(t - u < S^* \leq t)/u$. Then $\varphi^*(\lambda) = \lambda \int_0^\infty h(u)\, e^{-\lambda u}\, du$, and $\varphi^*(\infty) = h(0^+) = f_{S^*}(t)$ is the value at the constant atom. S^* is a constant plus a sum of independent Gamma variables, so its density is unimodal [3, Theorem 4, p. 11], proved in [4, proof of Theorem 4, pp. 194–195]; the constant shifts the mode only. Then $g(v) = f_{S^*}(t - v)$ for $v < t$ and $g(v) = 0$ for $v \geq t$ is unimodal on $[0, \infty)$, and so is its running average $h(u) = u^{-1} \int_0^u g$: before the mode of g, $g \geq h$; after it, $u\,(g(u) - h(u)) = u\, g(u) - \int_0^u g$ is nonincreasing. Since $h' = (g - h)/u$, h rises and then falls.

4. *Zero count.* For every constant κ', $h - \kappa'$ changes sign at most twice. The Laplace transform of a function with at most n sign changes has at most n zeros in $\lambda > 0$, counted with multiplicity: at a sign change u_1, multiply the transform by $e^{\lambda u_1}$; its derivative in λ is the transform of a function with one sign change fewer; Rolle's theorem and induction conclude. So $\varphi^* - \kappa' = \lambda \int_0^\infty (h - \kappa')\, e^{-\lambda u}\, du$ has at most two zeros in $\lambda > 0$. Also $\varphi^*(\lambda) \to 0$ as $\lambda \to 0^+$. Hence φ^* cannot rise above a level $\kappa' > 0$, fall below it and rise above it again: with $\varphi^*(0^+) = 0$ that would give three zeros. So φ^* is quasi-concave. It is analytic in $\lambda > 0$ and not constant, so its maximum over $\lambda \in [1, \infty]$ is attained at one point only: two maximisers would make φ^* constant between them.

5. *One term.* By (i) and (ii), an atom with $c \in (0,1)$ is a zero of $\varphi^* - \kappa^*$ of multiplicity at least 2, and an atom at $c = 1$ a zero of multiplicity at least 1; the constant atom is at $\lambda = \infty$. By step 4, S^* has at most two atoms of positive mean. As $K \geq 3$, (iii) applies, so κ^* is the maximum of φ^* over $[1, \infty]$, attained at every atom of positive mean. By step 4 there is only one such atom, and by step 1 it is not the constant. So $S^* = c^* \,\mathrm{Gamma}(m/c^*, 1)$ with $c^* \in (0,1]$.

Therefore $\max_{\mathcal{C}_K(m)} \Pr(S > t) = \max_{c \in (0,1]} Q(m/c, t/c)$ for every $K \geq 3$, and Eq. (10) follows from step 1. □

Lemma 0.8 (Monotone in the mean)**.** *$W(m,t)$ is nondecreasing in $m \in (0,t)$.*

Proof. Let $m < m' < t$ and let the law of S be in $\mathcal{C}(m)$. Then the law of $S + (m' - m)$ is in $\mathcal{C}(m')$, and $\Pr(S + m' - m > t) \geq \Pr(S > t)$. Therefore $W(m,t) \leq W(m',t)$. □

Reliability row

The cap. For $A \in [0,t)$ let $w(A) = \max_{c \in (0,1]} Q(A/c,\, t/c)$; it is continuous and nondecreasing in A, because $Q(a,x)$ increases in a. The *sharp cap* $A_1(t,\varepsilon)$ is the largest $A \in [0,t)$ with $w(A) \leq \varepsilon$, computed as described below. The row, on every day $k = 1, \ldots, T$, is

$$\mu_k \;\leq\; \frac{A_1(\beta^* \tau,\, \varepsilon)}{\beta^*}. \tag{11}$$

The row is linear: it is built in its exact form, and a value of `reliability_impl` at the cell is ignored (Section **??**).

Corollary 0.9 (The row is safe)**.** *Let the tracked mean satisfy Eq.* (11) *on day* k*. Then* $\Pr(D_k > \tau) \le \varepsilon$*.*

Proof. Let $m = \beta^* \mathbb{E}D_k$ and $A = A_1(\beta^*\tau, \varepsilon)$. The tracked mean bounds the true mean from above (Section **??**), so $m \le \beta^*\mu_k \le A < t$. If $m = 0$, then $D_k = 0$. Otherwise

$$\Pr(D_k > \tau) = \Pr(\beta^* D_k > t) \le W(m, t) \le W(A, t) = w(A) \le \varepsilon,$$

by Lemma 0.5, Lemma 0.8, Theorem 0.7 and the choice of A. □

Computation of A_1 (offline, once per cell, before the model is built):

1. $t = \beta^*\tau$. $\hat{A}$ is the root of $Q(\hat{A}, t) = \varepsilon$, found by bisection on $\hat{A}$ (Q increases in its first argument; `gammaincinv` solves for the second argument and does not apply).
2. $w(A)$: the function $c \mapsto Q(A/c, t/c)$ tends to 0 as $c \to 0$ and has a single peak on $(0, 1]$ (checked numerically for t from 10^{-3} to 100 and A/t from 0.05 to 0.999). Its maximum is found on a logarithmic grid of at least 400 values of c in $[\min(10^{-4}, t/100), 1]$, then refined by a golden-section search between the neighbours of the best grid point. If the best grid point is the smallest value of c, the grid is extended downwards.
3. If $\hat{A} < t$ and $w(\hat{A}) \le \varepsilon$, then $A_1 = \hat{A}$. Otherwise A_1 is found by bisection on A in $[\varepsilon t, \min(\hat{A}, t))$, to a relative tolerance of 10^{-9}. The lower end satisfies $w(\varepsilon t) \le \varepsilon$, by Markov's inequality $Q(a, x) \le a/x$. The bisection returns the lower end of its last interval, so $w(A_1) \le \varepsilon$ holds at the returned value.

Remark (the sharp cap and the exact one-rate cap). $\hat{A}$ is the exact cap of a single Gamma law of rate β^*, and $A_1 \le \hat{A}$, because $c = 1$ is in the maximum. Let $t_1(\varepsilon)$ be the value of t at which $\partial_c Q(\hat{A}/c, t/c)$ at $c = 1$ changes sign: $t_1 = 0.436$, 0.448 and 0.614 for $\varepsilon = 0.001$, 0.01 and 0.1. For $t \ge t_1$, $A_1 = \hat{A}$: on a grid of t in $[t_1, 1000]$, no $c < 1$ gives a larger tail than $c = 1$. Below t_1, $A_1 < \hat{A}$. For $\varepsilon = 0.01$:

$t = \beta^*\tau$	0.01	0.1	0.3	1	10
$A_1/\hat{A}$	0.142	0.642	0.961	1	1

When every term has the same rate and no repair occurs, the true state is a single Gamma law and $\hat{A}$ is exact for it. With several rates, or with repairs, the true state is a weighted sum, and below t_1 the exact one-rate cap is not safe. Example: $\varepsilon = 0.01$ and $\beta^*\tau = 0.1$, so $\hat{A} = 0.0055012$. A new part flies one mission of mean $\hat{A}/\beta^*$ and rate $4.56\,\beta^*$. Its mean meets the exact one-rate cap, but its tail is $1.565\,\varepsilon$. The sharp cap is $A_1 = 0.642\,\hat{A}$ there.

Remark (below the threshold). With $r = A/t$ and $s = t/c$, $Q(A/c, t/c) = Q(rs, s)$, and $c \in (0, 1]$ means $s \ge t$. For $t \le t_1(\varepsilon)$ the peak of $s \mapsto Q(rs, s)$ at the cap lies at $s \ge t$, so the maximum does not depend on t, and $A_1(t, \varepsilon) = r^*(\varepsilon)\,t$. The mean cap is then $A_1/\beta^* = r^*(\varepsilon)\,\tau$, whatever the rates: $r^* = 0.00355$, 0.0353 and 0.331 for $\varepsilon = 0.001$, 0.01 and 0.1.

Remark (the choice of β^**).* A larger t shrinks the range $s \ge t$, so $A_1(t, \varepsilon)/t$ is nondecreasing in t, and the mean cap $A_1(\beta^*\tau, \varepsilon)/\beta^* = \tau\, A_1(t, \varepsilon)/t$ is nondecreasing in β^*. Lemma 0.5 holds for every β^* at most the smallest rate. So the smallest rate gives the largest safe cap of this form.

Remark (conservatism). The class $\mathcal{C}(m)$ also contains laws that no schedule produces. Below t_1 the row can therefore reject some safe schedules. The row keeps each mission's own mean; β^* only bounds how dispersed each term may be.

Facts used by the encoding and loop arguments. (M1) The day map (8) is nondecreasing in (μ, g^μ). (M2) The true mean follows the map (8) exactly. (M3) If the tracked mean satisfies Eq. (11) and the true mean is not larger, then $\Pr(D_k > \tau) \le \varepsilon$ (Corollary 0.9). The loop row of Section **??** covers μ and, under ARD_1, g^μ.

Rainflow Model

This section applies to the cells (i, ℓ) with `model` = `rainflow`. Inside it, the subscript $i\ell$ is dropped from the per-cell parameters and from the states; k is the day.

The rainflow model is *distribution-free*. The law of a mission increment is measured from recorded load histories by rainflow counting (Palmgren–Miner damage), and it enters the model only through summary statistics: its mean, and, as the bound requires, its variance, the width of its range, or its cumulant generating function (CGF) at a fixed point s. No parametric family is assumed. The reliability requirement $\Pr(D_k > \tau) \le \varepsilon$ is certified by one of five tail bounds, chosen per cell by `bound_method`: `markov`, `cantelli`, `hoeffding`, `bernstein`, `chernoff`.

Modelling assumptions (normative).

- **A0 (non-negative damage).** Every increment, the initial damage and every replacement draw are non-negative.
- **A1 (independence).** The mission increments are mutually independent, and independent of the initial damage and of every replacement draw. Each replacement installs a new draw, independent of everything else. Every bound except Markov needs A1.
- **A2 (bounded ranges; Hoeffding and Bernstein).** Each mission increment lies in an interval of width b^{inc}_{jk} (input `support`). The initial damage is a known part plus independent terms, each in an interval; R_0 is the sum of their squared widths (input `R_0`) and b^0 the largest width (input `support_0`). The replacement draw is described in the same way by R^{new} and b^{new} (inputs `R_new`, `support_new`). A known damage has $R = b = 0$, the defaults.
- **A3 (repetition).** The row of the cell's bound holds on every day $k = 1, \dots, T$, and after day T the operating phase repeats with period H_2, schedule and increment profile included. This is the setting of the loop argument of Section **??**.

Width of a range. The input `support` of an increment X is the width b of an interval $[l, l + b]$ that contains X almost surely. Hoeffding's inequality uses the widths. Bernstein's inequality needs an upper bound of the centred term, and the width is one: $X - \mathbb{E}X \le l + b - l = b$, since $\mathbb{E}X \ge l$. A width b is consistent with a variance v only if $v \le b^2/4$ (Popoviciu's inequality: $\operatorname{Var} X \le \mathbb{E}(X - l - b/2)^2 \le (b/2)^2$); the check is in Section **??**.

Tracked states per bound. Every cell tracks the mean μ. Each bound adds at most one state, and under ARD_1 one anchor per tracked state (g^μ and g^v, g^R or g^K). Both laws are accepted for every bound.

`bound_method`	States	Increment inputs	Initial; replacement inputs	Forms
`markov`	μ	`mu`	–	`exact`
`cantelli`	μ, v	`mu`, `v`	`v_0`; `replacement_v`	all three
`hoeffding`	μ, R	`mu`, `support`	`R_0`; `R_new`	all three
`bernstein`	μ, v	`mu`, `v`, `support`	`v_0`, `support_0`; `replacement_v`, `support_new`	all three
`chernoff`	μ, K	`mu`, `cgf`, `s_chernoff`	`cgf_0`; `cgf_new`	`exact`

The column "Forms" lists the forms in which the row can be built; "all three" means `exact`, `tangent` and `pwl`. A linear row is built in its exact form, and a value of `reliability_impl` at its cell is ignored. R is the *Hoeffding budget*, the sum of the squared widths of the terms of the damage. K is the CGF of the damage at the cell's parameter s = `s_chernoff` > 0. The day increments are $\delta\mu_k = \sum_{j=1}^{M} x_{ijk}\, \mu^{\mathrm{inc}}_{jk}$, $\delta v_k = \sum_j x_{ijk}\, v^{\mathrm{inc}}_{jk}$, $\delta R_k = \sum_j x_{ijk}\, (b^{\mathrm{inc}}_{jk})^2$ and $\delta K_k = \sum_j x_{ijk}\, K^{\mathrm{inc}}_{jk}$, with $K^{\mathrm{inc}}_{jk} = \ln \mathbb{E}\, e^{sX_{jk}}$ the CGF of the increment at s (input `cgf`).

Day maps

Between replacements the damage is $D_k = \sum_t w_t X_t$: the X_t are the known part, the independent terms of the initial damage or of the replacement draw, and the mission increments; each weight $w_t \in (0, 1]$ is a product of factors $1-\rho$. Under ARD_1, write $D^- = G + \Delta$, with G the damage just after the last intervention and Δ the weighted increments added since. G and Δ are independent by A1, and a repair gives $D^+ = G + (1-\rho)\,\Delta$. Before the first intervention, G is the part of the initial damage that the starting anchors describe (Section **??**).

For a damage $D = \sum_t w_t X_t$, write $m(D) = \mathbb{E}D$, $V(D) = \operatorname{Var} D$, $R(D) = \sum_t w_t^2 r_t$ with r_t the squared width of X_t, and $K(D) = \ln \mathbb{E}\, e^{sD}$.

Proposition 0.10 (Day maps of the statistics)**.** *Let $\theta \in \{m, V, R\}$ with factor $a_\theta = 1-\rho$ for m and $a_\theta = (1-\rho)^2$ for V and R. Under A1:*

1. *a mission day adds the increment's statistic: $\theta(D+X) = \theta(D) + \theta(X)$ and $K(D+X) = K(D) + K(X)$;*
2. *ARD_∞: $\theta\big((1-\rho)D\big) = a_\theta\,\theta(D)$ and $K\big((1-\rho)D\big) \le (1-\rho)\,K(D)$;*
3. *ARD_1: $\theta(D^+) = a_\theta\,\theta(D^-) + (1-a_\theta)\,\theta(G)$ and $K(D^+) \le (1-\rho)\,K(D^-) + \rho\,K(G)$.*

Proof. Item 1: the mean and R are additive by definition, the variance and the CGF by independence. Item 2: m and V scale by $1-\rho$ and $(1-\rho)^2$; every weight is multiplied by $1-\rho$, so R scales by $(1-\rho)^2$. A CGF is convex with value 0 at 0, so $K\big((1-\rho)D\big) = K_D\big((1-\rho)s\big) \le (1-\rho)\,K_D(s) + \rho\,K_D(0) = (1-\rho)\,K(D)$, with K_D the CGF of D as a function of its argument. Item 3: by independence, $\theta(D^-) = \theta(G) + \theta(\Delta)$; for R the weights of the terms of G are unchanged and those of Δ are multiplied by $1-\rho$. Hence

$$\theta(D^+) = \theta(G) + a_\theta\,\theta(\Delta) = \theta(G) + a_\theta\big(\theta(D^-) - \theta(G)\big) = a_\theta\,\theta(D^-) + (1-a_\theta)\,\theta(G),$$

and, with item 2 applied to Δ,

$$K(D^+) = K(G) + K_\Delta\big((1-\rho)s\big) \le K(G) + (1-\rho)\big(K(D^-) - K(G)\big) = (1-\rho)\,K(D^-) + \rho\,K(G).$$

□

So each tracked state follows the generic map of Section **??** with the factor $a = 1-\rho$ for μ and K, and $a = (1-\rho)^2$ for v and R. Its anchor is its value just after the last intervention. The maps of μ, v and R are exact; the map of K is an upper bound, which is what the domination argument of Section **??** needs.

Starting and replacement values

The initial and replacement values of μ and v come from the inputs (Section **??**). Those of R and K are:

- R_0 from `R_0` and R^{new} from `R_new`, default 0;
- K_0 from `cgf_0`, default $s\,\mu_0$, and K^{new} from `cgf_new`, default $s\,\mu^{\text{new}}$.

With `replacement_as_new` all replacement values are 0 (Section **??**).

Why the defaults. By Jensen's inequality, $K(D) = \ln \mathbb{E}\, e^{sD} \ge s\,\mathbb{E}D$, with equality exactly when D is almost surely constant. So $s\mu_0$ is the exact CGF of a known initial damage μ_0, and a lower bound for a random one; a random initial damage needs its own CGF in `cgf_0`. The input checks of Section **??** require `cgf` $\ge s\,\mu^{\text{inc}}$ for every increment, $K_0 \ge s\mu_0$ and $K^{\text{new}} \ge s\mu^{\text{new}}$. A known damage has width 0, so $R_0 = 0$ is exact for it, and a random one needs `R_0`. The same holds for the replacement draw. A starting value of K below $s\mu_0$ makes the row optimistic. Example: $\tau = 1$, $\varepsilon = 0.1$, $s = 5$, a known initial damage $\mu_0 = 0.6$, and one mission that adds 0 or 0.5

with probability 1/2 each, so $K^{\text{inc}} = \ln(0.5 + 0.5\, e^{2.5}) = 1.886$. Then $\Pr(D > \tau) = 0.5$. The row with $K_0 = s\mu_0 = 3$ gives $4.886 - 5 > \ln 0.1$ and rejects the plan; a start at $K_0 = 0$ would give $1.886 - 5 = -3.11 \le \ln 0.1 = -2.30$ and accept it.

Rows of the dynamics

The rows are the generic rows of Section **??**, built for μ and for the one extra state of the bound, with the data below. $V^{\text{C}} = \varepsilon\tau^2/(1-\varepsilon)$ and $V^{\text{B}} = \tau(\tau - 2c)/(2\kappa)$ are the variance ranges of the Cantelli and Bernstein rows, with $\kappa = \ln(1/\varepsilon)$ and c from Eq. (15). Each natural range is the largest value that the bound's exact row admits at $\mu \ge 0$. The big-M rows need every initial value, starting anchor and replacement value inside the natural range of its state; the input checks of Section **??** require it.

State	Day increment	Factor a	Natural range	Initial value	Replacement value
μ; anchor g^{μ}	$\delta\mu_k$	$1-\rho$	τ	μ_0	μ^{new}
v; anchor g^{v}	δv_k	$(1-\rho)^2$	V^{C} or V^{B}	v_0	v^{new}
R; anchor g^{R}	δR_k	$(1-\rho)^2$	$2\tau^2/\kappa$	R_0	R^{new}
K; anchor g^{K}	δK_k	$1-\rho$	$s\tau - \kappa$	K_0	K^{new}

The anchors exist only under ARD_1, and the extra state only for its bound: v for Cantelli and Bernstein, R for Hoeffding, K for Chernoff.

Reliability rows

Write $d_k = \tau - \mu_k$. Let $(m, V, R^{\text{t}}, K^{\text{t}})$ be the true statistics of D_k. By Proposition 0.10 and Section **??**, each is at most the tracked state of the same name. Each row below is imposed on every day $k = 1, \dots, T$.

Markov (`markov`). By A0 and Markov's inequality, $\Pr(D_k > \tau) \le m/\tau \le \mu_k/\tau$. The row is

$$\mu_k \le \varepsilon\, \tau. \tag{12}$$

It uses the mean only and does not need A1 or A2.

Cantelli (`cantelli`). Cantelli's inequality gives $\Pr(D - m \ge a) \le V/(V + a^2)$ for $a > 0$. The row is

$$(1-\varepsilon)\, v_k \le \varepsilon\, (\tau - \mu_k)^2, \qquad \mu_k \le \tau, \tag{13}$$

equivalently $\mu_k + \kappa_C \sqrt{v_k} \le \tau$ with $\kappa_C = \sqrt{(1-\varepsilon)/\varepsilon}$. *Validity.* If $d_k > 0$,

$$\Pr(D_k > \tau) \le \Pr(D_k - m \ge \tau - m) \le \frac{V}{V + (\tau - m)^2} \le \frac{v_k}{v_k + d_k^2} \le \varepsilon,$$

because $v/(v + d^2)$ increases in v and decreases in d, and $\tau - m \ge d_k$. If $d_k = 0$, the row gives $v_k = 0$, so $V = 0$ and $D_k = m \le \tau$.

Hoeffding (`hoeffding`). $D_k - m$ is a sum of independent centred terms $w_t(X_t - \mathbb{E}X_t)$ with ranges of widths $w_t\sqrt{r_t}$, and the sum of their squares is R^{t}. Hoeffding's inequality gives $\Pr(D_k - m \ge a) \le \exp(-2a^2/R^{\text{t}})$, and $\exp(-2d^2/R) \le \varepsilon$ exactly when $d^2 \ge (\kappa/2)\, R$. The row is

$$(\tau - \mu_k)^2 \ge \frac{\kappa}{2}\, R_k, \qquad \mu_k \le \tau. \tag{14}$$

Validity. If $R^{\text{t}} = 0$, every term is constant and $D_k = m \le \tau$; this covers $d_k = 0$, where the row gives $R_k = 0$. Otherwise $d_k > 0$, and $\Pr(D_k > \tau) \le \exp\bigl(-2(\tau - m)^2/R^{\text{t}}\bigr) \le \exp(-2d_k^2/R_k) \le \varepsilon$, because $\exp(-2d^2/R)$ increases in R and decreases in d, and $\tau - m \ge d_k$.

Bernstein (`bernstein`). The *support constant* of the cell is

$$b = \max\Big(b^0,\ b^{\text{new}},\ \max_{j,k} b^{\text{inc}}_{jk}\Big), \qquad c = \frac{\kappa\, b}{3}, \tag{15}$$

over the missions j and the days k of both phases. It is a constant, not a decision variable. Every weighted term satisfies $w_t(X_t - \mathbb{E}X_t) \le w_t\, b_t \le b$, whatever the schedule, because $w_t \le 1$. Bernstein's inequality for independent terms bounded above by b gives $\Pr(D - m \ge a) \le \exp\big(-a^2/(2(V + b\,a/3))\big)$ for $a > 0$; the right side is at most ε exactly when $a^2 - 2c\,a - 2\kappa V \ge 0$. The row is

$$(\tau - c - \mu_k)^2 \ge c^2 + 2\kappa\, v_k, \qquad \mu_k \le \tau - c, \tag{16}$$

equivalently $\mu_k + c + \sqrt{c^2 + 2\kappa v_k} \le \tau$. With $v_k \ge 0$, the two rows give $d_k \ge 2c$. *Validity.* $a^2/(2(V + b\,a/3))$ increases in $a > 0$ and decreases in V, and $\tau - m \ge d_k \ge 2c$; so $\Pr(D_k > \tau) \le \varepsilon$ when $b > 0$. When $b = 0$, every term is constant and $D_k = m \le \tau$.

Remark (the side row). The quadratic row alone also admits the point $d_k = 0$, $v_k = 0$: a known damage equal to τ. That point is safe, but it is cut off from the rest of the feasible set. The side row $\mu_k \le \tau - c$ removes it and keeps the branch $d_k \ge c + \sqrt{c^2 + 2\kappa v_k}$. At $\mu = v = 0$ the row needs $\tau \ge 2c$; a positive variance needs $\tau > 2\kappa b/3$, the input check of Section **??**.

Chernoff (`chernoff`). For $s > 0$, Markov's inequality applied to e^{sD} gives $\Pr(D_k \ge \tau) \le \exp(K^{\mathrm{t}} - s\tau)$. The row is

$$K_k - s\,\tau \le \ln \varepsilon. \tag{17}$$

Validity. $K^{\mathrm{t}} \le K_k$, so $\Pr(D_k > \tau) \le \exp(K_k - s\tau) \le \varepsilon$. The row bounds K by $s\tau - \kappa$, its natural range; $K \ge 0$, because $K(X) \ge s\,\mathbb{E}X \ge 0$ by Jensen's inequality and A0. A known zero damage satisfies the row only if $s\tau \ge \kappa$ (input check $s > \ln(1/\varepsilon)/\tau$, Section **??**). A replacement day satisfies it only if $K^{\text{new}} \le s\tau - \kappa$; the input checks require every replacement state to satisfy the cell's row.

Linear forms. Markov and Chernoff rows are linear: they are built in their exact form, and a value of `reliability_impl` at the cell is ignored. The Cantelli, Hoeffding and Bernstein rows have the form $\chi_k \le f(\mu_k)$ with f convex in μ, and accept the forms `exact`, `tangent` and `pwl` of Section **??**. Every form keeps the side row.

Bound	χ	$f(\mu)$	Side row	$f(0)$
Cantelli	v	$\varepsilon(\tau - \mu)^2/(1 - \varepsilon)$	$\mu \le \tau$	$\varepsilon\tau^2/(1 - \varepsilon)$
Hoeffding	R	$2(\tau - \mu)^2/\kappa$	$\mu \le \tau$	$2\tau^2/\kappa$
Bernstein	v	$\big((\tau - \mu)^2 - 2c(\tau - \mu)\big)/(2\kappa)$	$\mu \le \tau - c$	$\tau(\tau - 2c)/(2\kappa)$

Each f is a parabola in μ with a positive leading coefficient, so it is convex. For Bernstein with $b > 0$, the tangent point must satisfy $\mu_r < \tau - c$, an input check of Section **??**. At $\mu_r \ge \tau - c$ the tangent has a nonnegative slope and lies below f, so at every $\mu \le \tau - c$ it is at most $f(\tau - c) = -c^2/(2\kappa) < 0$, and no point with $v \ge 0$ satisfies the row.

Optional row. If `v_max_user` is given for a Cantelli or Bernstein cell, the row $v_k \le V^{\text{user}}$ is added on every day.

Facts used by the encoding and loop arguments. (M1) Every day map is nondecreasing in each state and anchor: its coefficients a and $1 - a$ lie in $[0, 1]$. (M2) The true statistics follow the maps with equality for μ, v and R, and with $\le$ for K (Proposition 0.10). (M3) If the tracked states satisfy the row of the cell's bound and the true statistics are not larger, then $\Pr(D_k > \tau) \le \varepsilon$ (the validity paragraphs above). The loop row of Section **??** covers μ, the extra state of the bound and, under ARD_1, their anchors.

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